

IoT-Driven Early Warning System for Diabetic Foot Ulcer Deploying ML Algorithm

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Abstract—Patients with long-term diabetes and poor glycemic index often develop peripheral neurovascular disorder. This neurovascular complication often results in Diabetic Foot Ulcer (DFU). Approximately 40 to 50% of patients are affected by foot ulceration who are diagnosed with diabetes. It is analyzed that approximately 20% of hospitalization among patients concerned to diabetes mellitus is due to foot ulceration. Hence, early detection and prevention of diabetic foot ulcers are essential. This paper focuses on early detection of chronic foot lesions such as blisters, fasciitis, and callus in diabetic patients by monitoring foot pressure. This project aims to develop a cost-efficient, portable foot pressure smart insole and check its dependability and credibility to prevent foot ulceration. Hardware is set up to monitor foot pressure in diabetic patients continuously via a flexi-force sensor. This is accomplished by placing pressure sensors at five pressure points in a smart insole which is then used by the patient, and made user-friendly by transmitting and storing the results on a server. The proposed Internet of Things (IoT)-based smart insole makes it easier to evaluate the patient's feet every day at home, enabling prompt detection of early signs and severity tracking. The goal of real-time ulcer detection system is to improve proactive diabetic treatment, decrease amputations, and avoid complications by incorporating Machine Learning (ML) algorithms. The proposed system indicates the patient's current status, which can be accessed by both the patient as well as the doctor through a mobile application developed on NetBeans.

Keywords – Diabetic Foot Ulcer, Internet of Things, Machine Learning, Flexi-force Sensor, KNN Algorithm

I. INTRODUCTION

One of the main consequences of diabetes is considered to be diabetic foot ulcers. Blood artery constriction, which results in poor blood circulation to the foot, is one of the vascular issues that can arise from elevated blood sugar levels for any patient is said to be affected by DFU. More than 50 million people in the world have diabetes and prevalence is among those over 45- 65 years of age [1]. Diabetes often leads to complications affecting the extremities, particularly the feet. One common issue is the development of foot ulcers, which can occur when a person experiences diminished sensation in their lower limbs. This condition, known as peripheral neuropathy, is frequently

observed in individuals managing diabetes over extended periods [2]. Lack of sensation in the foot and over-pressure on the foot primes to prolonged micro trauma, decline in underlying tissues, and eventually foot ulceration. Also, infections can develop without a warning. As a result, the patient's foot develops gangrene and loses feeling which leads to further complications [3]. Although peripheral neuropathy usually affects the sensory nerves that sense pain or temperature, it may also have an effect on the motor nerves that control muscular contraction. Figure 1 depicts the key feature of preventing Foot Ulcers.



Figure 1. Key preventions of Foot Ulcer

Generally DFU is classified according to their nature and severity into three types namely; Neuropathic DFU, Ischemic DFU and Neuroischemic DFU [4]. 1) Neuropathic DFU - Peripheral neuropathy, which damages the nerves in the foot, is the cause of neuropathic DFUs. Due to the absence of feeling in the feet caused by this ailment, people find it difficult to perceive pain or injuries. Usually, pressure sites like the soles of the feet are the site of neuropathic DFUs. 2) Ischemic DFU - Reduced blood flow to a specific location is referred to as ischaemia. Peripheral Artery

Disease (PAD), which is frequently observed in diabetics, causes insufficient blood flow to the feet, leading to ischaemic DFU. Arteries which provide blood to the feet constrict or get blocked as a result of PAD, which lowers the flow of oxygen and nutrients. Insufficient blood flow hinders the healing of wounds and can result in the formation of ulcers, typically on the heels, toe tips, or other high-pressure locations. 3) Neuroischemic DFU - It is a hybrid of chemic and neuropathic DFU. It happens when peripheral artery disease impairs blood flow to the foot in a person with diabetic neuropathy. In this instance, ulcers develop as a result of both nerve injury and a decreased blood flow. Although it can affect other parts of the foot, neuroischemic DFUs frequently occur on pressure points.

Smartwatches that seamlessly monitor vital signs are the best examples of how the Internet of Things is revolutionising the healthcare industry. [5 & 6]. By combining health data, these platforms allow for real-time comparison to predetermined standards. This study examines the revolutionary possibility of IoT and ML in primary foot ulcer diagnosis as IoT usage in healthcare soars [7 & 8]. To promote proactive diabetic care, the emphasis is on simplifying constant monitoring and undertaking issues in the healthcare industry, such as disguised unemployment. By bridging the gap between daily life and medical intervention, the wearable prototype such as smartwatches is a potential step towards proactive, easily available care. As this work unfolds, it should clarify the process, design complexities, and modular architecture that support the creation of the prototype. This research imagines a future for early identification is commonplace, reducing the influence of foot ulcers and changing the diabetic care scenarios through the visualization of IoT and ML.

Further the paper is divided and organized as following sections: motivation, background and a detailed literature survey is presented in Section II. Section III contracts the proposed approach of IoT-Driven early warning system for diabetic foot ulcer deploying ML algorithm. Section IV includes the results and discussion of proposed system along with discussion. Finally, this paper concludes the proposed work in Section V with its future scope.

II. MOTIVATION AND BACKGROUND

A. Problem statement and Motivation

Diabetic foot ulcers pose a serious risk because it hinders wound healing and provide an environment that is conducive to infection due to variables like poor blood circulation, high blood sugar, and weakened skin. If a person has diabetes-related neuropathy and/or poor circulation, the patient is more likely to have foot problems and consequences. Given the seriousness of DFUs, this work develops a novel strategy for early detection that uses machine learning and the internet of things. Figure 2 depicts the statistics of the prevalence of diabetic foot affected patients globally in 2024.

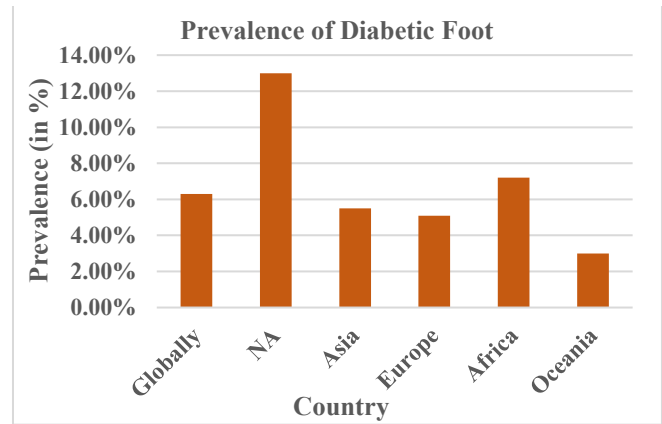


Figure 2. Prevalence of Diabetic Foot

B. Literature Survey

A groundbreaking Egyptian study has shed light on the critical issue of diabetic foot ulcers and their impact on patient outcomes. Researchers at Cairo University Hospital conducted a comprehensive case-control study, involving 200 participants, to explore the potential of artificial intelligence in predicting these ulcers [9]. By analyzing 19 key factors from patients' medical histories and foot imaging, the team developed two predictive models: an artificial neural network and a decision tree. The neural network emerged as the superior tool, boasting an impressive 97% accuracy rate in automated ulcer prediction. This research not only highlights the power of AI in medical diagnostics but also emphasizes the importance of implementing preventive strategies in diabetic care clinics to combat this serious complication.

P. N. Thotad et al., proposed a DL based approach which aims to offer researchers a comprehensive overview of the current landscape in automatic detection of diabetic foot ulcer [10]. It highlights the necessity of employing both traditional machine learning and advanced Deep Learning (DL) techniques for rapid and reliable diagnostic decision-making by clinicians. While traditional ML leverages image features for accurate DFU identification, advanced DL, particularly Convolution Neural Network (CNN)-based solutions, emerges as more promising [11]. The article serves as a valuable resource for researchers, facilitating a clear understanding of the DFU identification task's nuances and guiding the formulation of future research goals in this domain.

The realm of diabetic foot ulcer prevention has been revolutionized by recent advancements in wearable technology and sensor miniaturization [12]. This study explored the potential of deep neural networks in automating the classification of diabetic foot images. The research focused on distinguishing between normal and abnormal foot conditions using various convolutional neural network models. Among these, the EfficientNet model stood out, achieving remarkable accuracy with a 96.86% overall success rate, 97.8% precision, 96.3% recall, and a 96.5% f1-score. The findings underscore the promising role of advanced neural networks in enhancing DFU detection and management, potentially transforming diabetic foot care practices.

Gaetano et al., introduces a novel non-invasive, photonic-based device designed to aid in the control of diabetic foot ulcers, defined as a dangerous side effect of diabetes that can cause severe impairments, such as amputation of the foot, or even death [13]. Unlike current methods, which often rely on

invasive biopsies, this device utilizes Hyper-Spectral Imaging (HSI) and thermal imaging technologies to assess ulcer conditions. These imaging techniques help measure key biomarkers such as oxy-hemoglobin (hbO2) and deoxy-hemoglobin (hb), allowing for the calculation of Peripheral Oxygen Saturation (SpO2) and Tissue Oxygen Saturation (StO2). The proposed device is of i) home version: for patients and 2) professional version: for healthcare providers. This version includes advanced photonic tools like tunable diodes for more precise diagnosis and monitoring. Additionally, the device incorporates signal processing algorithms for noise reduction and enhances pixel-level accuracy through super-resolution techniques.

One of the most prevalent side effects of diabetes is DFUs. It's critical to identify infection, ischaemia, and lesions in DFU in order to examine ulcers and design treatments [14]. Deep learning-based computerised segmentation of DFU infection and ischaemia has recently demonstrated encouraging results. Deep Neural Networks (CNN), particularly CNN, are used in the majority of state-of-the-art DFU image classification techniques to extract differentiated features. Fully connected neural networks then use the recovered features to forecast class discrimination probabilities.

The prediction is dependent on a single input image during testing, yet trained parameters and characteristics for incremental learning are still used explicitly [15]. This work suggests Class Knowledge Banks (CKBs), which are made up of trainable units that can efficiently extract and represent class knowledge, as a way to make better use of the knowledge in the training data. In CKB, each unit is utilised to calculate how similar its presentation is to that of an input image, the legitimacy can be defined as the average similarity between the units in the CKB and their presentation. In this manner, the class knowledge taken from the training data and saved in the CKBs in addition to the input images and trained parameters.

Diabetic foot syndrome is a complex condition influenced by several key factors, including peripheral neuropathy, peripheral arterial disease, and infection. One of the challenges in managing this condition is its often-subtle onset, with early symptoms frequently going unnoticed. In recent years, the incidence of diabetic foot syndrome has risen, becoming a significant complication of diabetes that adversely affects patients' quality of life [16]. Given the intricacies involved in the condition, Artificial Intelligence (AI) has shown potential in areas such as early detection and risk assessment for foot ulcers or even amputations. Using sensor-based technologies, AI can aid in timely intervention. This overview highlights notable findings from research in this area, focusing on both the AI methodologies employed and the clinical outcomes observed.

S. Gupta et al., has extensively evaluate deep learning object detection networks for diabetic foot ulcer detection, acknowledging the potential in automatically localizing ulcers [17]. However, challenges include numerous false positives and difficulty discriminating ulcers from other skin conditions. Introducing a second classifier with a negative dataset may address this, but the feasibility of obtaining all possible negative examples poses challenges. Additionally, this approach may impact network size and complexity, potentially slowing down inference speed. An alternative solution involves segmenting the foot from its surroundings to simplify the environment for trained models, mitigating the need to account for complex objects.

III. PROPOSED IoT BASED SMART INSOLE

The proposed IoT-based smart insole for diabetic foot ulcer warning system has been implemented by deploying ML algorithm. The basic block diagram of the proposed work is illustrated in Figure 3. It consists of three modules namely i) Force Sensing Resistor (FSR) integrated with Arduino NodeMCU ESP8266 TPX00165 module, ii) implementation of Machine Learning model, and iii) patient alert and notification module. The five-foot regions for implementing the proposed architecture are identified as shown in Figure 4, which bear the foremost pressure and are subject to the most noteworthy strengths amid standing. Based on the pressure applied the FSR will send the resistance to the Arduino and it is continuously stored in data storage with the cloud platform. Sensor measures the pressure values at these pressure points for further decision process. The proposed method chooses Arduino platform since it is open-source electronics platform built on user-friendly hardware and software. Arduino boards have the ability to take mechanical inputs from sensors and process them based on the algorithm developed and embedded on it to take better decisions out of it. The ease of K Nearest Neighbour classification is one of its advantages. The user simply needs to decide on two things: (1) how many neighbours K, and (2) what distance measure to use. For instance, City-block distance, Euclidean distance, and Mahalanobis distance are real-time examples.

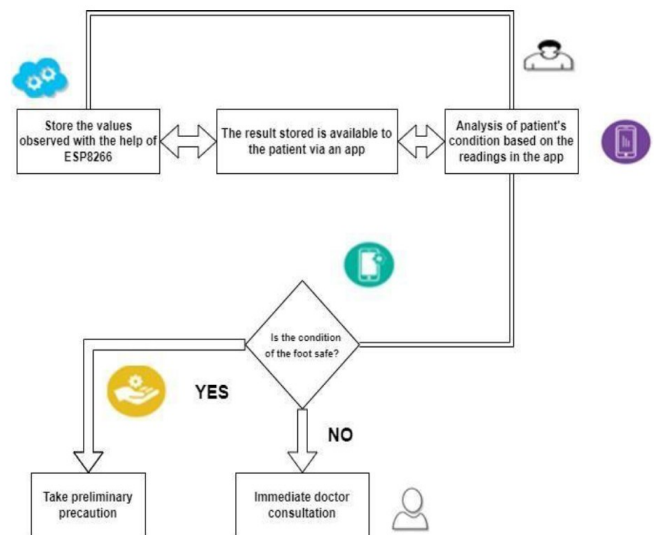


Figure 3. Block diagram of a proposed model

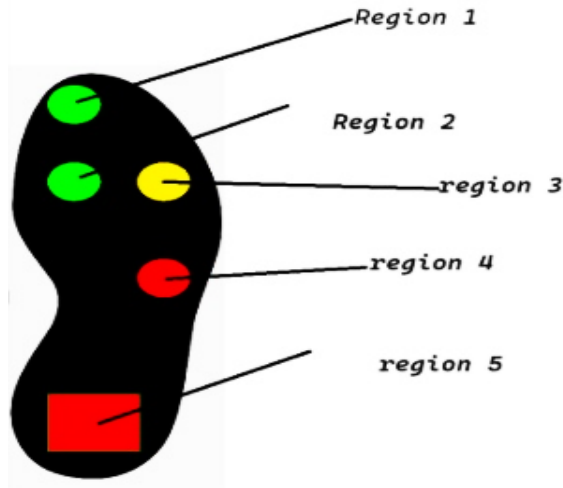


Figure 4. Foot regions considered for developing the proposed smart insole

The proposed prototype consists of basic five pressure points namely; heel, ball of foot, toe, arch and lateral side of the foot whose pressure ranges are measured by the sensors. Table 1 illustrates the five different areas concentrated in this work, to measure the pressure with the help of FSR that are mapped to the risk level of determining diabetic foot ulcer. These values helps to identify the risk level of the patient who wears the smart insole. The threshold values for indicating the safer diabetic level will be stored in the cloud whose values are compared with the sensor measured values for further decision making with the help of ML algorithm.

Table 1. FSR pressure vs risk level for different foot regions

FSR Location	Pressure Range (g)	Risk Level	Remarks
Heel (FSR)	100-200	Low Risk	Normal weight-bearing area with low stress
Ball of Foot (FSR)	200-400	Medium Risk	Moderate pressure, needs monitoring
Toe (FSR)	> 400	High pressure zone, Risk of tissue damage	High pressure zone, risk of tissue damage
Arch (FSR)	< 100	Low Risk	Minimal pressure, no risk in this area
Lateral Side (FSR)	100-300	Medium Risk	Higher pressure than normal, needs attention

K Nearest Neighbor (KNN) algorithm has been used to the trained model for predicting the foot state. Foot ulcers are classified into two categories: low and high. The collected data is monitored continuously for several weeks and the states for the foot can be checked through the Android-based mobile application developed for this prototype. Patient alerts and notifications are implemented in the developed mobile applications that sends

notification to the guardians who support preventive care, and timely intervention for the patient health.

A. Integration of FSR with Arduino

Force Sensor Resistors are lightweight, lean, and cost-effective that detects force or pressure associated with its surface which is connected to five diverse points of the foot insole as shown in Figure 5. Node MCU, an open-source platform built on the ESP8266, is used to design the proposed prototype, to facilitate device connections and data transfer via the Wi-Fi protocol. With an inbuilt ESP8266 (Wi-Fi Module) and incredibly low power requirements, this microcontroller was designed especially for IoT applications.

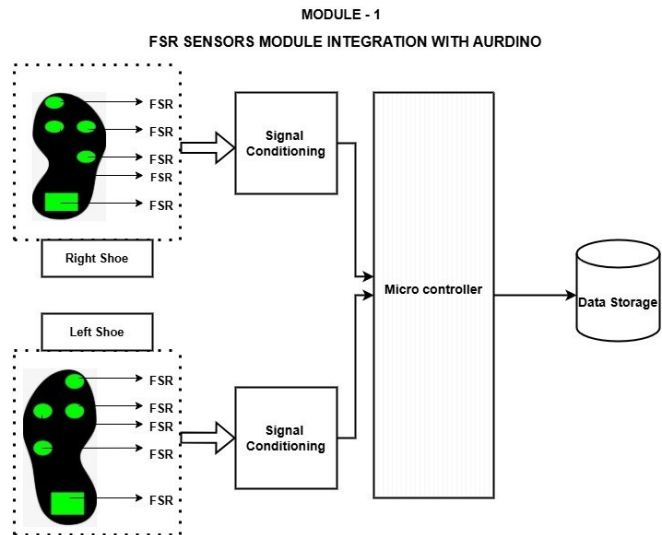


Figure 5. Integration of FSR with Arduino

FSR produces the electrical resistance based on the amount of pressure exerted on it; greater force results in lower resistance. The integration of FSRs with the Arduino NodeMCU ESP8266 entails incorporating components like Analog-to-Digital Converter (ADC) for converting the analog signal (i.e., pressure) measured from FSR and converts it into digital signals for further processing. The multiple reading of FSR from five different regions of the foot is stored for further processing. Reading ADC Values from FSR using Arduino involves the steps which are shown in Table 2.

Table 2. Connection establishment between FSR and Arduino NodeMCU

No	Steps	Connection Establishment
1	Initialize ADC	Set up and configure the Arduino ADC to read multiple samples from FSR.
2	Sample FSR Input	Read the ADC values corresponding to the FSR input. Store the multiple samples.
3	Define Pressure Thresholds	Fix threshold values to differentiate between normal and abnormal pressure levels. State the higher and lower boundaries for normal pressure based on FSR readings.

4	Trigger Alerts	Compare the filtered ADC values against the defined thresholds.
		If abnormal pressure is detected, trigger an alert or initiate further analysis.

B. Machine Learning Model

A non-parametric, supervised learning classifier named KNN classifies forecasts how to cluster individual data points according to their closeness. Finding the closest data point or neighbors from a training dataset for a query is how the KNN algorithm works. The closest distances from the query point are used to determine the closest pressure values (Figure 6). It uses a majority voting rule to determine which class occurred the most after identifying the K closest data points. (i.e., low = 2 and high = 1). The final classification for the query is determined by the class that showed up the most. While some KNN variations prioritize overall classification and accuracy, others concentrate on determining the ideal k values. The confusion matrix, provides a tabular summary of the model's predictions is used to calculate the accuracy of the model. By calculating the counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) accuracy is predicted through Equation 1.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \quad (1)$$

C. Alert and Notification Module

This module facilitates data transmission as shown in Figure 7 between the smart shoe systems with external devices (i.e., Smart Phone). The predicted values are transferred to the smartphone via a Wi-Fi (ESP8266) module connected to the smart insole. The mobile application is developed to view the visual representation of smart shoes with FSR color values indicating the various conditions of pressure states. This application displays the data in real-time and stores it for later analysis. The automatic vibrator is installed to automatically turn ON/OFF driven on the pressure values. The DC motor converts electrical energy into mechanical motion is integrated to vibrators that can alert users to specific pressure points and help in the improvement of an individual's walking pattern promoting a more balanced and natural walking pattern.

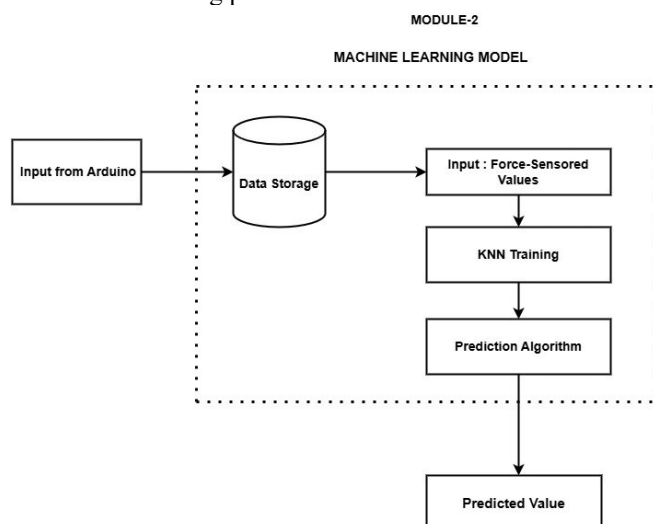


Figure 6. Proposed Machine Learning Model

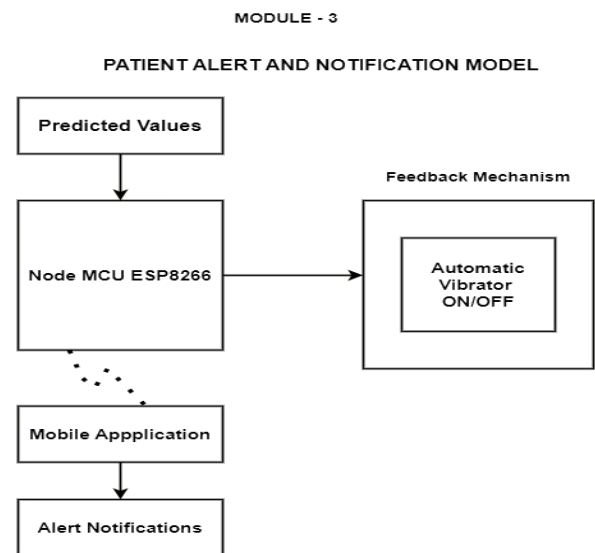


Figure 7. Data transmission by Alert and Notification Module

IV. RESULTS AND DISCUSSION

The proposed IoT-based smart insole has been developed as a prototype by embedding five FSR sensors at the defined pressure points which records the values periodically and send to the cloud (Figure 8). The recorded analog signals are converted into digital data, which has been recorded and stored as multiple samples fed to the central processor of the board; NodeMCU. This developed prototype has been integrated with Arduino NodeMCU ESP8266 through connecting the respective pins of the board with the sensors, as illustrated in Figure 9.

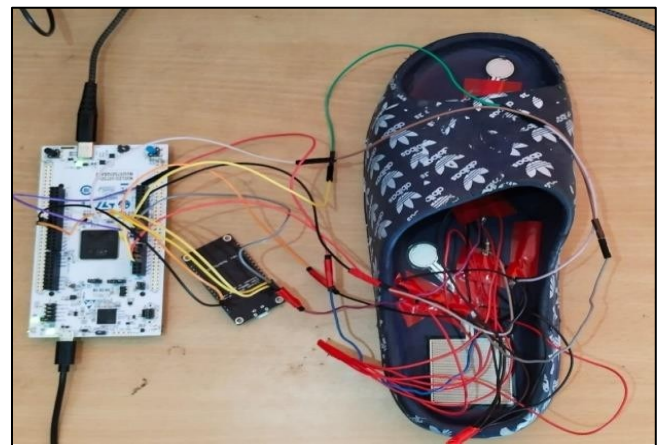


Figure 8. Prototype of the proposed smart insole

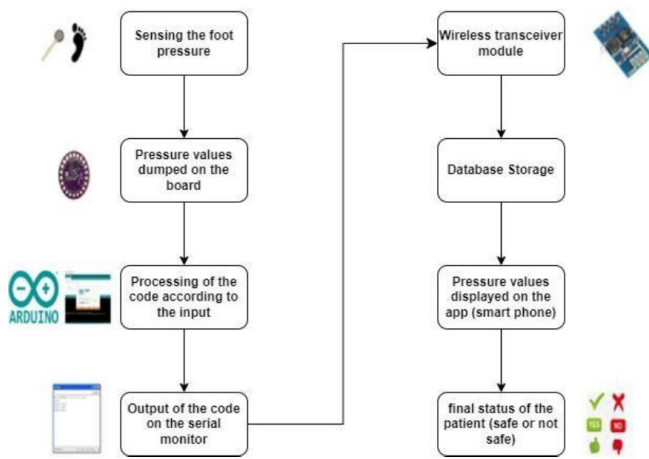


Figure 9. Architecture of the proposed prototype

FSR values will be recorded as illustrated in Figure 10 where; “#273#0#0#769” denotes the readings sensed by the sensors FSR1, FSR2, FSR3, FSR4, and FSR5. Multiple samples are collected from FSR and stored in data storage, after executing KNN algorithm it shows risk factor by comparing with FSR pressure table for various foot regions. These values will be sent to mobile application developed using NetBean platform. The snippet of developed Arduino code has been shown in Figure 11. The recorded values will be sent to the cloud through processing by KNN algorithm developed through Python platform. Then the recorded values are compared with the threshold values to check the risk factor of diabetes. A visual representation of risk levels has been mentioned for a better understanding of the results where; Red represents high risk, Yellow represents moderate risk and Green indicated safer range (Figure 12). Nearly 181 times the model has been trained to check the accuracy whose sample results are shown in Figure 13, where FSR 1 shown 181 times low, FSR 2 shown 178 low and 3 high, FSR 3 shown 160 low, 18 medium and 3 high, FSR 4 shown 169 low, 3 medium and 9 high and FSR 5 shown 167 low and 14 high.

The proposed ML model and the results has been validated by improving precision, recall and F-measure metrics. Hyperparameter settings can be changed for the proposed model. Performance of the model is frequently enhanced by fine-tuning these parameters periodically. For instance, the model's precision may be enhanced by raising its regularisation strength, whilst its recall may be enhanced by decreasing it. As discussed the accuracy is measured through confusion metrics, once it is analysed the next step to improve accuracy is to fine-tune the model. This can be done with the help of adjusting the model's classification threshold. Similarly, reliability of the model can be improved through two methods namely; (1) Validation: To make sure the model is accurate and dependable; it is tested on fresh data. It aids in locating any mistakes or shortcomings in the model and (2) Regularisation: To avoid overfitting, this technique entails introducing extra constraints to the model. Finally, the performance of proposed ML algorithm can be enhanced by increasing the amount of training data because it will have more examples to learn from the data. This is especially helpful if the existing training dataset is unbalanced or small.

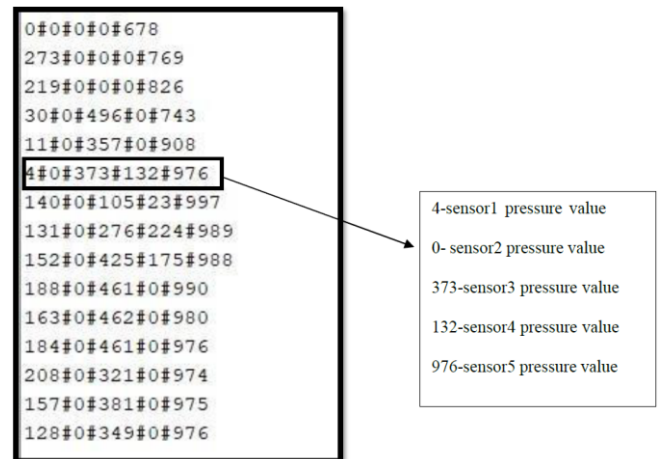


Figure 10. FSR recorded output values

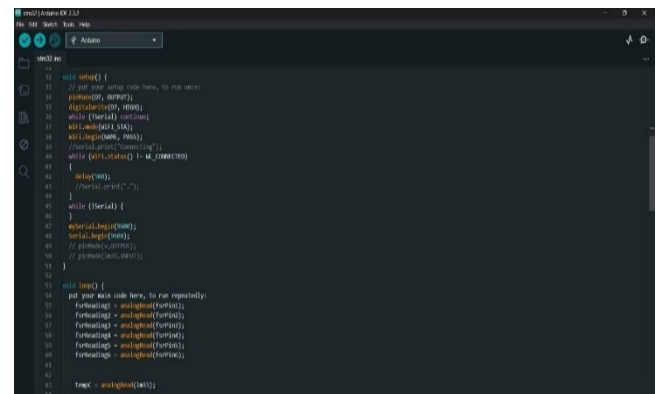


Figure 11. Snippet of the developed Arduino code

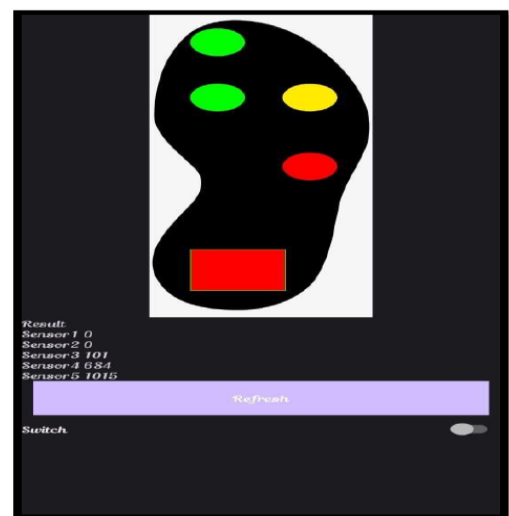


Figure 12. Snippet of the developed Arduino code

The proposed KNN algorithm has been evaluated by calculating metrics such as Precision, Recall, F-Measure and Accuracy. The evaluated metrics has been compared with the various algorithms/methodologies like EfficientNet, Neural Networks and Photonic-based approach as discussed in literature survey. The measured metrics has been illustrated in Table 3, where the proposed KNN algorithm receives 98.26% of accuracy compared to the state-of-art where the scores of EfficientNet, Neural Networks and Photonic-based approach measures 96.86%, 97.6% and 95.9% respectively.

Table 3. Comparison study of various algorithms

Algorithm	Precision	Recall	F-Measure	Accuracy
EfficientNet	97.8%	96.3%	96.5%	96.86%
Neural Networks	98.5%	97.2%	97.2%	97.6%
Photonic-based	96.1%	95.8%	95.8%	95.9%
Proposed KNN algorithm	98.8%	98.1%	98.1%	98.26%



Figure 13. Virtual outputs of trained models shown in the mobile app

Thus, using the proposed smart insole allows patients and healthcare providers to access the data through a smartphone application, providing real-time information about the patient's foot health status. By regularly wearing these sensor-equipped shoe and monitoring the provided readings, patients can promptly detect any abnormalities and seek medical advice if necessary. Early intervention can prevent the progression of potential ulcers, leading to better outcomes for patients. This approach empowers patients to take proactive measures in managing their foot health and enables timely medical intervention, ultimately reducing the risk of diabetic foot complications.

V. CONCLUSION AND FUTURE WORK

The proposed IoT -based smart insole for predicting diabetic foot ulcers leverages smart insole technology, offering a solution for early detection and prevention of complications in diabetic patients. Real-time monitoring via sensor smart insoles, prompt and affordable intervention capabilities, effective sensor data processing, and system design are some of the salient aspects. Using K-Nearest Neighbours (KNN) classification, it has achieved a

noteworthy 98.26% accuracy rate, indicating great dependability in predicting the formation of ulcers. The strength of the work is that it achieves maximum accuracy which can be improved by improving Precision, F-Measure, Recall values and training the model with more datasets so that it can be highly validated. Whereas, it faces issues in developing ML algorithm for KNN classification such as computational complexity, data dependency, interpretation, overfitting and underfitting. The future enhancements include ideas like sensor addition such as Integration of humidity and temperature sensors, Improved monitoring of additional physiological factors. And also the personalized care tailored predictions based on long-term patient data, customized intervention strategies. By combining advanced technology, this project paves the way for more effective, personalized diabetic foot care, potentially revolutionizing the field of preventive healthcare for diabetic patients.

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